MILITARY SPECIFICATION

QUADRANT, GUNNER'S: MIAIEl

1. SCOPE

1.1 This specification covers one type of gunner's guadrant which measures angles of cant and elevation. The level vial in this quadrant is radioactively illuminated.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U.S. Army, Frankford Arsenal

D6510977	Elevation Test Fixture Assembly
D8292946	Adapter Level Test Quadrant, Gunner's Ml
A8658940	Inspection Aid
F11731007	Quadrant, Gunner's: MIAIEl

PACKAGING DATA SHEET

11731007	Packaging of Quadrant, Gunner's: MIAIEl

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(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

DEFENSE SUPPLY AGENCY MANUAL

DSAM 4145.8 Radioactive commodities in the DOD Supply Systems

(Copies of DSAM 4145.8 may be obtained from the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C. 20402.)

3. REQUIREMENTS

3.1 Regulations pertaining to radioactive devices.- Marking, identification, handling, and shipping of radioactive devices shall be in accordance with DSAM 4145.8.

3.2 Fabrication.- The quadrant shall be manufactured in accordance with Drawing F11731007.

3.3 General specification.- The following requirements of MIL-F-13926 shall be met:

a. Order of precedence.
b. Dimensions and tolerances.
c. Inorganic protective surface finishes.
d. Part identification and marking.
e. Workmanship.

3.4 Environmental.

3.4.1 Storage temperatures.- The quadrant shall show no evidence of physical failure due to being thermally stabilized at -80° and +160° Fahrenheit (F).

3.4.2 Operating temperatures.- The quadrant shall meet the applicable requirements of 3.8 while thermally stabilized at -40° and +150°F.

3.5 Index alinement.- With the micrometer knob indicating zero, the zero index on Level Holder 11730281 shall coincide with the index on Arm 6129938 within the width of the index line on the arm. This requirement shall be met on each side of the quadrant.

3.6 Accuracy.- The quadrant shall indicate angular values from

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zero to 1600 roils (with respect to a horizontal plane) within 0.40 mil.

3.7 Torque.- The running torque required to rotate the micrometer- . knob shall not exceed 2.0 pound-inches at the operating temperatures specified in 3.4.2.

3.8 Level vial.- The level vial bubble shall be visible in all ambient light conditions

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three gunner's quadrants plus three (3) each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample

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and a record of this inspection, including certificates of conformance for materials shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected. in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP'S the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables 1, II and 111 herein. Examination and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I

CLASSIFICATION	REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE		
MAJOR: AQL 1.0% Defective		
101. Index alinement	3.5	4.6.3
102. Accuracy	3.6	4.6.4
103. Torque	3.7	4.6.5
104. Level vial visibility	3.8	4.6.6
MINOR: NONE		

NOTE: The tests in Table I shall be conducted at a standard ambient temperature of between 60° and 90°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

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4.4 Special sampling.

4.4.1 General.- Three quadrants shall be selected at random as a special sampling from each 100 produced. The samples shall meet the requirements and tests in Table II.

TABLE II

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURES
301.	Regulations pertaining to radioactive devices	3.1	Certification
302.	Fabrication	3.2	Drawings - Visual
303.	General specification	3.3	MIL-F-13926 - Visual

NOTE : The tests in Table II shall be conducted at a standard ambient temperature of between 60° and 90°F.

4.4.2 Environmental tests.- Three quadrants shall be selected at random from each 50 produced, or from each month's production, whichever occurs first. Each sample shall meet the requirements and tests in Table III and shall then meet the requirements in Table I.

TABLE 111

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
304.	Storage temperatures	3.4.1	4.6.1
305.	Low temperature operation (-40°F)	3.4.2	4.6.2
306.	High temperature operation (+150°F)	3.4.2	4.6.2

4.4.3 Failure of sample.- Should any item of a special sampling fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

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4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Inspection Documents when provided with the Technical Data Package (TDP).

4.5.2.2 Contractor design.- The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the "Component Inspection" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming- Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment design requirements.- Special testing equipment to conduct the test for the accuracy requirements shall conform to Drawings D6510977 "Elevation Test Fixture Assembly", D8292946 "Adapter Level Test Quadrant, Gunner's Ml", and A8658940 "Inspection Aid". Level the test fixture by means of the inspection aid positioned on the trunnion cross bar. Assemble the test adapter to the fixture and level the adapter with the inspection aid positioned on the referenced surface of the test adapter. In this position, the vernier scale of the elevation fixture shall be set to zero.

4.6 Test methods and procedures.

-4.6.1 Storage temperature.- The gunner's quadrant shall be exposed to the temperature conditions specified in 3.4.1, utilizing testing equipment in accordance with the "Test Facilities" requirements of MIL-F-13926. The quadrant shall be placed in the test chamber and the temperature of the chamber reduced gradually to -80°F and shall remain at this temperature for a minimum of 4 hours. At the completion of this 4 hour period, the temperature shall then be gradually raised to +160°F and the quadrant shall remain at this temperature for a minimum of 4 hours. At the completion of this 4 hour period the temperature of the test chamber shall be gradually reduced to room ambient temperature (+60° to +90°F) at which time the quadrant. shall be removed from the test chamber. The quadrant shall then be subjected to a visual and tactile examination to determine that there is no evidence of distortion, cracks, level vial breakage, or undue iorregularities of moving parts. Caution shall be exercised during this test to avoid subjecting the chest assemblies to thermal shock.

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NOTE : The rate of temperature change in the test chamber shall not exceed 20°F per hour throughout the temperature cycling tests of 4.6.1 and 4.6.2.

4.6.2 Operating temperature.- The quadrant shall be exposed to the -- temperature conditions specified in 3.4.2, utilizing testing equipment in accordance with the "Test Facilities" requirements of MIL-F-13926. The quadrant shall be placed in the cold chamber and the temperature of the chamber reduced gradually (see NOTE) to -40°F. The quadrant shall remain at this temperature for a minimum of 5 hours. At the completion of this five hour period, and while at -65°F the quadrant shall be subjected to the test in 4.6.6. Upon completion of this portion of the test, the quadrant temperature shall be gradually raised to room ambient temperature. The quadrant shall then be placed into an oven chamber and the temperature shall be raised gradually to +150°F. The quadrant shall remain at this temperature for a period of 5 hours. At the completion of this 5 hour period, and while at +150°F the quadrant shall again be subjected to the test in 4.6.6. Upon completion of this test, the temperature of the test chamber shall be gradually reduced to room ambient temperature (+60° to +90°F) at which time the quadrant shall be removed from the test chamber. Caution shall be exercised during this test to avoid subjecting the quadrant to thermal shock.

4.6.3 Index alinement.- Index alinement at the settings specified in 3.5 shall be determined by a visual and tactile examination. As a means to facilitate the measuring of any misalinement of the indices a commercial type optical comparator with appropriate tolerance mark may be utilized.

4.6.4 Accuracy.- The adjusted testing equipment as specified in 4.5.2.3 shall be utilized for this test. Set the calibrated inspection aid to zero roils elevation and position it on the seat provided on the test adapter, and observe that the level vial bubble is centralized Set the quadrant under test to zero roils elevation and place it on the seat provided on the test adapter in the same position as the inspection aid. Now observe the position of the level vial bubble of the unit under test, it shall be centralized and in coincidence with the level vial setting of the inspection aid. If the level vial setting of the quadrant under test does not coincide with the inspection aid, adjustment shall be made to obtain coincidence by rotating the micrometer of the unit under test. The amount of error read from the micrometer scale shall not exceed the tolerance specified in 3.6. In addition, accuracy shall be tested at a minimum of 4 diversified angles of elevation for each scale of the quadrant. This is accomplished by checking the quadrant under test with the settings of the inspection aid and following the procedure out-lined above as required for each scale sitting. Any error, not cumulative, shall be recorded and shall not exceed the tolerance specified in 3.6.

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4.6.5 Torque. - A commercial type torque wrench capable of measuring to the torque values specified in 3.7 shall be utilized for this test. An appropriate adapter to accommodate the micrometer knob shall be used with the torque measuring device. Readings taken with the torque wrench shall be taken in both directions and while in motion. The running torque required to rotate the micrometer knob shall conform to the requirement of 3.7 when tested at the extreme operating temperatures of 3.4.2.

4.6.6 Level vials.- Place the quadrant in a dark room having a controlled light source. The observer and the quadrant under test must become adapted to the dark. After being adapted to the darkness,the observer shall visually examine the level vial bubble. It must be visible to assure conformance with 3.8. Gradually raise the ambient level of light in the dark **room one** foot **candle of illumination in the area of the level** vial. The level vial bubble shall be visible to assure conformance with 3.8.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking.- Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 11731007. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The MIAIEl Gunner's Quadrant is primarily intended for use in tank and artillery gunnery for measuring angles of cant and elevation.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification
b. Selection of applicable levels of preservation, packaging and packing (see 5.1)
c. Applicable packaging data sheet number (see 5.1)
d. Applicable stock number
e. Provisions for First Article testing

6.3 Radioactive material.- Prospective bidders should be cognizant of the need for an Atomic Energy Commission license covering the procurement and handling of radioactive material.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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